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Assignment 5(2012)(24-25 Second)

Fondamental concepts of math (The University of Hong Kong)

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A5/MATH2012/2024-25/2nd

THE UNIVERSITY OF HONG KONG
DEPARTMENT OF MATHEMATICS

MATH2012 Fundamental Concepts of Mathematics

Assignment 5

Due date : Apr 24, 2025 before 12:00 noon.

Instructions

- Please scan and submit your work as a **PDF** file in **moodle**. No late work will be accepted.

  Warning: Make sure that your work is properly scanned. **Oversized, blurred or upside-down images will NOT be graded**.

Guidelines on Writing

- You should always give precise and adequate explanations to support your conclusions. Note that being able to write clearly is as important a mathematical skill as being able to solve equations. Putting an idea on paper requires careful thought and attention. Writing forces you to express your ideas clearly and in the proper order.
- Proper writing of mathematics may not have been emphasized in your previous mathematics courses, but as you start learning university-level mathematics, you may have to adjust your approach to doing mathematics.
- Suggestions:
  - Write in complete sentences. Do not just give the solution in fragmentary bits and pieces. Clarity of presentation of your argument counts.
  - Explain the meaning of every symbol that you introduce and avoid starting a sentence with a symbol.
  - Use the implication symbol "$\Longrightarrow$" and the "if and only if" symbol "$\Longleftrightarrow$" correctly.
  - Be careful to use quantifiers "for all", "for any", "for some", "there exists" correctly. You cannot expect the reader to understand which you mean if you omit these quantifiers.
  - Avoid using the symbols "$\because$" and "$\therefore$". Use "because/since", "therefore/hence/it follows that".

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References on Mathematical Writing

- Chapter 0 of *Mathematical Proofs: A Transition to Advanced Mathematics*
- Section I (Chapter 2 - 4) of *How to Think Like a Mathematician: A Companion to Undergraduate Mathematics*
- Y.M. Chan, *Writing Mathematics*

Remark

- You are welcome to contact the instructor or the tutor(s) if you have any difficulties. See '*Course Information*' for availabilities.

You need only to give your answers to those questions indicated with *. Other problems are optional and will not be marked. However, we do expect you to eventually finish all problems.

Questions with * in this assignment:

Q.3, Q.4, Q.6, Q.7, Q.10, Q.14, Q.15

Chapter 5 (Section 5.1 to 5.3)

1. Let $A = \{1, 2, 3\}$ and $B = \{1, 2\}$. For each of the following, list the required functions as subsets of $A \times A$, $A \times B$ or $B \times A$.
   (a) Write down all injective functions from A to A.
   (b) Write down all surjective functions from A to A.
   (c) Write down all injective functions from A to B.
   (d) Write down all surjective functions from A to B.
   (e) Write down all injective functions from B to A.
   (f) Write down all surjective functions from B to A.

2. Let $f : \mathbb{R}^2 \longrightarrow \mathbb{R}$ be defined by
$$f(x, y) = \frac{2x + 1}{5 + y^2}.$$
   Prove or disprove that f is surjective.

*3. Let A be the set of odd integers and B the set of even integers. Define a function $f : A \times B \longrightarrow A \times A$ by
$$f(a, b) = (3a - b,\ a + b).$$
   (a) Prove or disprove that f is injective; (b) Prove or disprove that f is surjective.

*4. Prove that for any nonempty set A, there exists an injective function $f : A \longrightarrow \mathcal{P}(A)$.

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5. Let A be a nonempty set. For any subset $X \subseteq A$, define the *characteristic function* $f_X : A \longrightarrow \{0, 1\}$ by

$$f_X(a) = \begin{cases} 1 & \text{if } a \in X, \\ 0 & \text{if } a \in A \setminus X. \end{cases}$$

For each of the following, show that $f_X \circ f_X$ is defined. Then find $f_X \circ f_X$ and prove your answer.

(a) $X = \mathbb{N}$ and $A = \mathbb{R}$.

(b) $X =$ the set of odd integers and $A = \mathbb{Z}$.

(c) $X =$ the set of even integers and $A = \mathbb{Z}$.

*6. Let $f : \mathbb{R} \longrightarrow \mathbb{R}$ be a function satisfying $f(x + f(y)) = f(x) - y$ for any $x, y \in \mathbb{R}$. Prove that f is surjective.

*7. Prove or disprove the following statements:

For any nonempty sets A, B, C and for any functions $f : A \longrightarrow B$ and $g : B \longrightarrow C$,

(a) if f is injective, then $g \circ f$ is injective.

(b) if g is surjective, then $g \circ f$ is surjective.

8. Prove or disprove the following statement:

For any nonempty sets A and B with $A \neq B$ and for any functions $f : A \longrightarrow B$ and $g : B \longrightarrow B$, if $g \circ f = f$, then $g = id_B$ where id_B is the identity function on B.

9. (a) **Disprove** the following statement:

For any nonempty sets A, B, C and for any functions $f : A \longrightarrow B$, $g : C \longrightarrow A$ and $h : C \longrightarrow A$,

$$\text{if } f \circ g = f \circ h, \text{ then } g = h.$$

(b) The statement in (a) will become true if an additional assumption is imposed on the function $f : A \longrightarrow B$. What is this assumption? **Prove** that the statement in (a), together with this additional assumption, is true.

*10. (a) **Disprove** the following statement:

For any nonempty sets A, B, C and for any functions $f : A \longrightarrow B$, $g : B \longrightarrow C$ and $h : B \longrightarrow C$, if $g \circ f = h \circ f$, then $g = h$.

(b) The statement in (a) will become true if an additional assumption is imposed on the function $f : A \longrightarrow B$. What is this assumption? **Prove** that the statement in (a), together with this additional assumption, is true.

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Chapter 5 (Section 5.4)

11. Let $f : A \longrightarrow B$ be a bijective function.

    (a) Show that $(f^{-1})^{-1} = f$.

    (b) Show that
    $$f^{-1} \circ f = id_A \quad \text{and} \quad f \circ f^{-1} = id_B$$
    where id_A and id_B are respectively the identity functions on A and B.
    (*Remark*: Use *Definition 5.8* for both (a) and (b).)

12. (a) Let $f : \mathbb{R} \setminus \{8\} \longrightarrow \mathbb{R}$ be defined by $f(x) = \dfrac{3x-4}{x-8}$. Show that f is injective but not surjective.

    (b) Find a set B so that $g : \mathbb{R} \setminus \{8\} \longrightarrow B$ given by $g(x) = \dfrac{3x-4}{x-8}$ is bijective. Determine $g^{-1}(x)$ for any $x \in B$.

13. Let S be a nonempty set and $\mathcal{P}(S)$ the power set of S. For any $A, B \in \mathcal{P}(S) \setminus \{\emptyset\}$, define a relation R on $\mathcal{P}(S) \setminus \{\emptyset\}$ by
    $$ARB \text{ if there exists a bijective function } f : A \longrightarrow B.$$
    Prove that R is an equivalence relation on $\mathcal{P}(S) \setminus \{\emptyset\}$.

*14. Let $f : A \longrightarrow B$ be a function. Suppose there is a function $g : B \longrightarrow A$ such that
    $$g \circ f = id_A \quad \text{and} \quad f \circ g = id_B,$$
    where id_A and id_B are respectively the identity functions on A and B.

    (a) Prove that f is bijective.

    (b) Show that the function g is equal to the inverse function f^{-1} of f.
    (*Remark*: Use *Definition 5.8*)

*15. Let $f : A \longrightarrow B$ and $g : B \longrightarrow C$ be two bijective functions. Use the result of **Q.14** (and without using *Corollary 5.27*), prove that $g \circ f : A \longrightarrow C$ is bijective and find the inverse function of $g \circ f$.

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